

AMCAS Assignment 1

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Part A – ngspice: 6T SRAM Read Margin

Central question: does the bitcell read correctly at 0.7 V and 85 °C?

Task	Result	Interpretation
1	$\Delta V(BL, BLB)$ at 2 ns = 729.86 mV; $q_{\max}$ = 205.52 mV.	Reads correctly; margin is $\sim 10.6\times$ the 69 mV lecture reference.
2	Access width 0.24 μm : ΔV = 852.34 mV, $q_{\max}$ = 271.22 mV.	Read disturb grows; cell is more marginal at worse PVT corners.
3	ΔV falls below 25 mV at $V_{DD} \approx 0.448$ V.	Within 1.1–0.6 V sweep, read still safe; failure starts below ~ 0.45 V.
4	At 85 °C: ΔV = 582.12 mV, $q_{\max}$ = 220.76 mV.	Limiting failure mode is read disturb, not sense-amp resolution.

Table 1: Part A summary of nominal, sizing, supply, and temperature results.

Task 1: Nominal Read Margin

The bitline differential at $t = 2$ ns is 729.86 mV ($q_{\max} = 205.52$ mV), giving $\sim 10.6\times$ the 69 mV sense-amp threshold. At nominal $V_{DD} = 1.1$ V the pull-down NMOS devices dominate the access transistors, keeping $v(Q)$ well below the inverter trip point.

Task 2: Access-Device Sizing

Widening access transistors to 0.24 μm increases ΔV to 852.34 mV but raises $q_{\max}$ by 31.9% to 271.22 mV, degrading the β -ratio and eroding read stability. At worst-case PVT corners this reduced SNM could cause a destructive read.

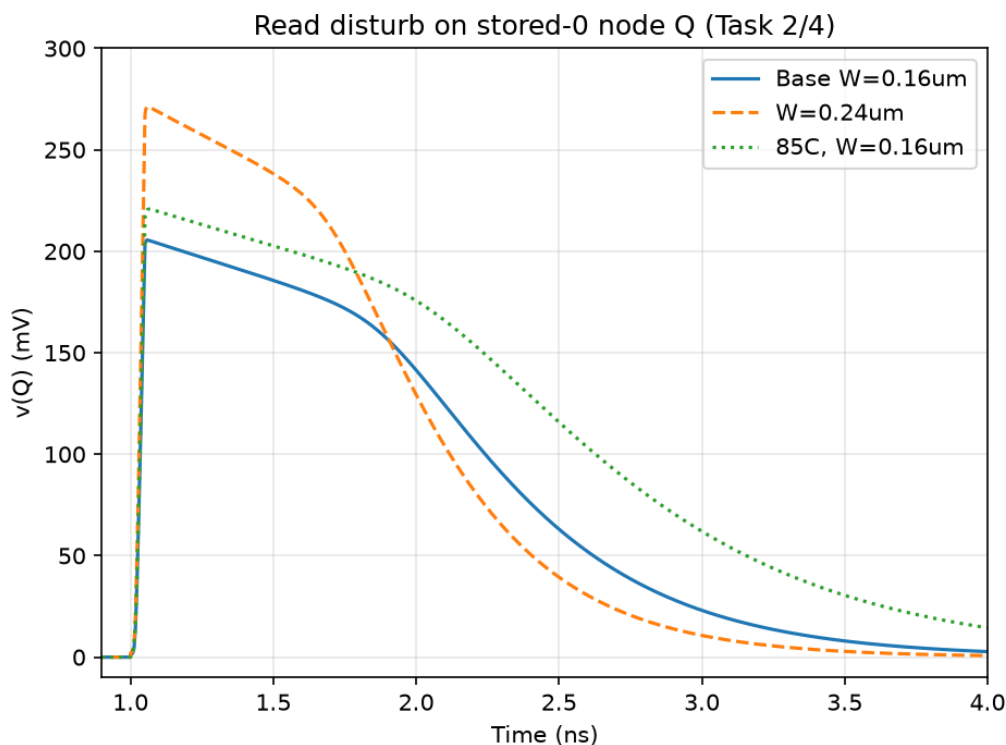


Figure 1: Read disturb $v(Q)$: nominal 205.5 mV, widened access 271.2 mV, 85 °C 220.8 mV.

Task 3: Supply Sweep

ΔV decreases monotonically with V_{DD} ; the 25 mV threshold is crossed at ≈ 0.448 V (interpolated). Within 1.1–0.6 V the minimum ΔV is 143.9 mV ($\sim 5.8\times$ above threshold), so the read is safe across this range.

V_{DD} (V)	1.10	1.05	1.00	0.95	0.90	0.85
ΔV (mV)	729.9	673.6	616.0	557.1	497.4	437.2
V_{DD} (V)	0.80	0.75	0.70	0.65	0.60	0.55
ΔV (mV)	376.8	316.6	257.1	199.1	143.9	94.2
V_{DD} (V)	0.50	0.45	0.40			
ΔV (mV)	53.6	25.7	10.7			

Table 2: Part A Task 3 – supply sweep. Interpolation between 0.45 V and 0.40 V gives the 25 mV crossing near 0.448 V.

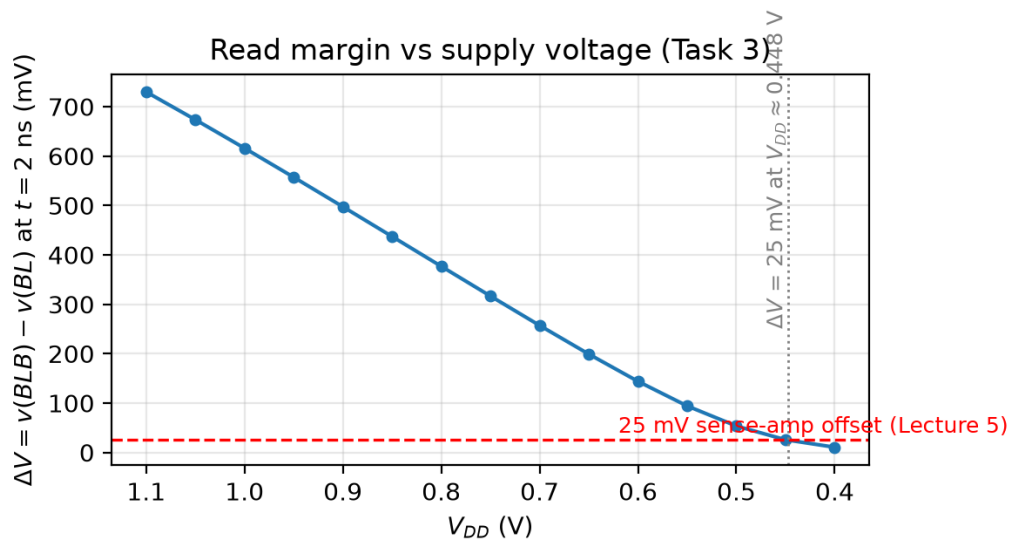


Figure 2: ΔV at 2 ns vs. V_{DD} ; 25 mV crossing near 0.448 V, safe across 1.1–0.6 V.

Task 4: Temperature Sensitivity

At 85 °C, ΔV drops 20.2% to 582.12 mV and $q_{\max}$ rises 7.6% to 220.76 mV. Reduced carrier mobility lowers drive current while increased subthreshold leakage raises the stored-node disturbance, making read disturb the dominant high-temperature failure mechanism.

Part B – CACTI: 2 MB SRAM L2

Central question: how fast, how big, and how leaky is a 2 MB SRAM at 45 nm?

Task 1: Baseline Organization

CACTI reports 2.90184 ns access time, 0.792885 nJ read energy, 562.632 mW leakage, and 11.4742 mm² area. The optimal organization is Ndwl/Ndbl/Nspd = 4/2/1, balancing decoder delay, bitline length, wordline RC, and H-tree routing.

Task 2: Capacity Scaling

Capacity	Access (ns)	Read energy (nJ)	Leakage (mW)	Area (mm ²)
256 kB	2.41714	0.658021	90.2876	4.96464
512 kB	2.48451	0.678967	158.126	5.94616
1 MB	2.63376	0.716643	293.680	7.59985
2 MB	2.90184	0.792885	562.632	11.4742
4 MB	3.53084	0.930134	1102.31	18.4579
8 MB	4.47269	1.21975	2183.82	39.4961
16 MB	6.36235	1.74213	4244.52	74.6677

Table 3: Part B Task 2 – capacity sweep (256 kB–16 MB).

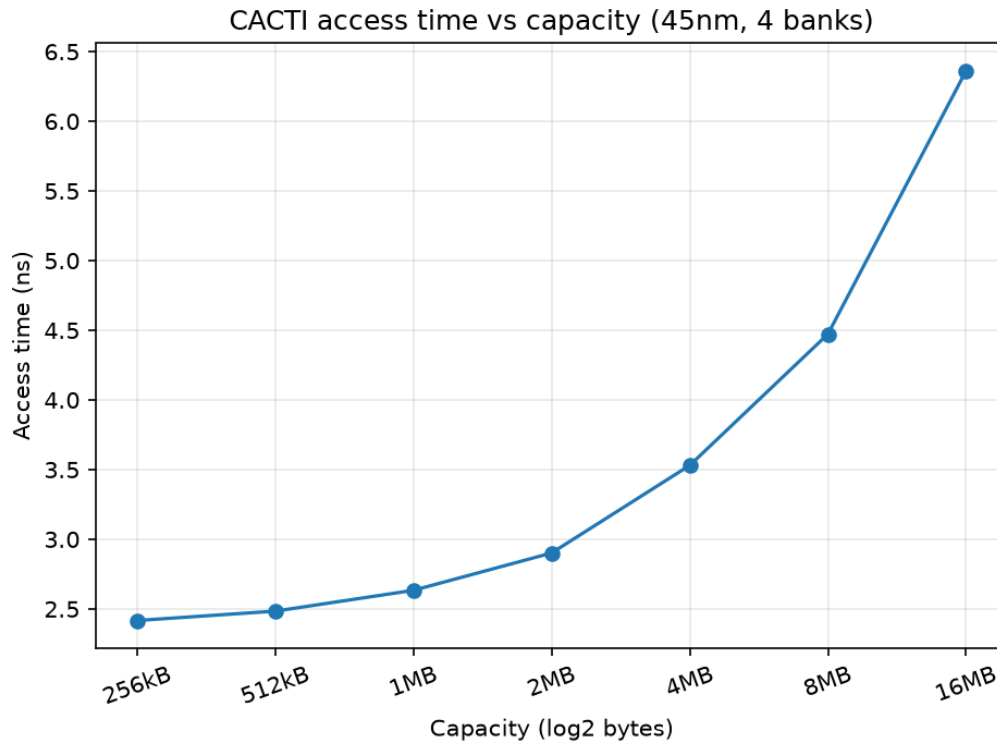


Figure 3: Access time vs. $\log_2(\text{capacity})$; linear to 2 MB, super-linear after (H-tree RC dominates).

Access time grows $2.63\times$ for $64\times$ capacity (256 kB→16 MB). Growth is super-linear at larger sizes because wire delay and H-tree routing dominate. Leakage and area scale near-linearly.

Task 3: Organization Objective

Objective	Weight	Access (ns)	Ndwl/Ndbl/Nspd	Interpretation
Pure delay	100:0:0:0:0	2.89231	4/2/1	More mat parallelism minimizes critical path.
Pure area	0:0:0:0:100	3.35518	2/2/1	Fewer subdivisions reduce peripheral overhead.
ED ² P	0:0:0:100:0	2.90184	4/2/1	Delay benefit dominates energy–delay trade-off.

Table 4: Part B Task 3 – objective comparison (weight order: delay, dynamic energy, leakage, cycle time, area).

Pure delay and ED²P both select 4/2/1; pure area selects 2/2/1 with a 16% access-time penalty. ED²P agreeing with pure delay confirms delay reduction outweighs extra peripheral energy at this node.

Task 4: Bitline Delay Check

CACTI reports a bitline delay of 0.4069 ns; the lecture RC formula yields 18.7 ns ($46\times$ overestimate). The discrepancy arises because the hand model assumes full-swing passive RC discharge, whereas the actual read is partial-swing (25–50 mV) terminated early by the sense amplifier, and CACTI uses hierarchical bitline segments.

Part C – NVSim: 2 MB STT-MRAM vs SRAM

Central question: same capacity, same periphery, different bitcell – what changes?

Metric	SRAM (CACTI)	STT-MRAM (NVSim)	Ratio
Area	11.474 mm ²	3.590 mm ²	0.31 $\times$
Read hit latency	2.902 ns	1.591 ns	0.55 $\times$
Write latency	2.902 ns	10.608 ns	3.66 $\times$
Read energy	0.793 nJ	0.763 nJ	0.96 $\times$
Write energy	0.851 nJ	0.435 nJ	0.51 $\times$
Leakage power	562.6 mW [†]	425 mW	0.76 $\times$

Table 5: Part C Task 1 – SRAM vs. STT-MRAM at 2 MB, 45 nm. [†]Cross-tool comparison (CACTI vs. NVSim). NVSim-internal SRAM reference: 6.393 mm², 0.85/0.52 ns, 3196 mW; against that, STT-MRAM ratios are 0.56 $\times$ area, 1.87 $\times$ /20.4 $\times$ read/write latency, 0.13 $\times$ leakage.

Task 1: SRAM–STT-MRAM Comparison

STT-MRAM delivers 0.31 $\times$ area (1T-1MTJ at $\sim 54 F^2$ vs. SRAM's 146 F^2) and 0.76 $\times$ leakage (magnetic storage needs no standby bias), but incurs a severe 3.66 $\times$ write penalty from spin-torque switching dynamics. These are cross-tool ratios; NVSim's own SRAM baseline confirms the expected 0.13 $\times$ leakage advantage.

Task 2: Strengths and Weaknesses

Config	Area (mm ²)	Rd lat. (ns)	Wr lat. (ns)	Rd E (nJ)	Wr E (nJ)	Leakage (mW)
SRAM baseline	6.393	0.850	0.520	0.575	0.531	3196.1
STT-MRAM	3.590	1.591	10.608	0.763	0.435	425.2
TMR high	3.580	1.596	10.657	0.761	0.433	425.2
Half reset current	3.590	1.571	10.588	0.762	0.433	425.0

Table 6: Part C Tasks 2–4 – NVSim variants at 2 MB, 45 nm.

Best STT-MRAM metrics: area (0.56 $\times$), leakage (0.13 $\times$), write energy (0.82 $\times$). Worst: read latency (1.87 $\times$), write latency (20.4 $\times$), read energy (1.33 $\times$).

Task 3: TMR Sensitivity

Raising TMR from 2:1 to 3:1 produces negligible array-level changes (largest: write latency +0.46%). TMR improves sense-amp signal margin and yield, not speed – an important distinction for MRAM scaling.

Task 4: Write-Current Sensitivity

Halving reset current (200 $\rightarrow$ 100 μ A) does not change cell area (MTJ footprint dominates at 54 F^2). Read latency drops -1.3% and write energy -0.5% . The primary STT-MRAM advantage is density and standby power, enabling $\sim 4\times$ more capacity in the same area – the key lever in Part E.

Part D – Ramulator 2: DDR4 Miss Stream

Central question: how do scheduler, address mapping, and channel count change latency and throughput?

Configuration	Cycles	Avg read lat.	Row-hit rate	Note
FRFCFS + RoBaRaCoCh	9 594	262.93	53.9%	Baseline
FCFS-proxy (buf=1)	42 937	61.90	69.6%	No reordering; much slower
ChRaBaRoCo	78 845	1844.21	55.9%	Row locality destroyed
2 channels	4 889	233.57	32.7%	Best throughput

Table 7: Part D – scheduler, address-mapping, and channel comparison (2048-request trace).

Task 1: Baseline Scheduler and Mapping

With FRFCFS and RoBaRaCoCh the trace completes in 9 594 cycles (262.93 avg read latency, 53.9% row-hit rate). FRFCFS prioritizes row-buffer hits, and RoBaRaCoCh maps consecutive addresses to columns within the same row.

Configuration	Cycles	Read lat.	Hits	Misses	Conflicts	Hit rate
FRFCFS + RoBaRaCoCh	9 594	262.93	1 079	215	707	53.9%
FCFS proxy	42 937	61.90	–	–	–	69.6%
ChRaBaRoCo	78 845	1844.21	1 121	7	878	55.9%
2 channels	4 889	233.57	326	143	528	32.7%

Table 8: Part D – row-buffer breakdown across configurations.

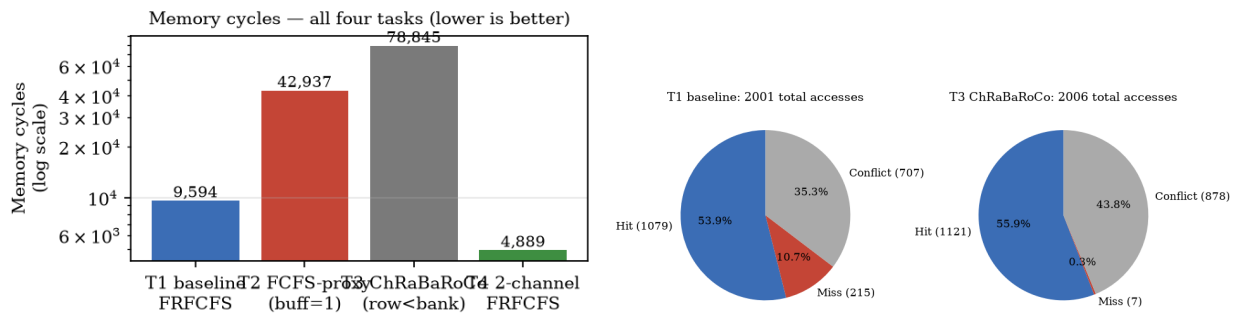


Figure 4: Total memory cycles (left); row hit/miss/conflict split for baseline (right).

Task 2: Scheduler Comparison

The FCFS-proxy achieves higher row-hit rate (69.6%) but $4.5\times$ worse throughput (42 937 cycles), demonstrating that bank-level parallelism and request reordering matter more than raw hit rate.

Task 3: Address Mapping

ChRaBaRoCo is catastrophic (78 845 cycles): placing row bits below bank bits maps consecutive addresses to different rows in the same bank, converting most accesses to costly conflicts (878 vs. 707 baseline).

Task 4: Channels and Timing Sensitivity

A second channel cuts cycles nearly in half (4 889) despite lower row-hit rate (32.7%).

Timing parameter	Cycles	Avg read lat.	Increase from baseline
Baseline	9 594	262.9	–
tRCD: 22 $\rightarrow$ 44	10 534	291.9	+9.8%
tRP: 22 $\rightarrow$ 44	10 625	281.9	+10.7%
tRAS: 52 $\rightarrow$ 104	10 003	269.0	+4.3%
tCCD: 4 $\rightarrow$ 8	12 203	322.1	+27.2%

Table 9: Part D Task 4 – timing-sensitivity experiment.

tCCD is the binding constraint (+27.2%), far exceeding tRCD/tRP/tRAS, because the 53.9% hit rate means many accesses are gated only by column-to-column spacing within an open row.

Part E – gem5: Whole-Program Simulation

Central question: does the program you care about actually run faster with STT-MRAM?

Part B gives SRAM a 7-cycle L2 hit latency (2.90 ns at $\sim$ 2 GHz); Part C shows the same die area holds $\sim$ 8 MB of STT-MRAM at 14-cycle hit latency. gem5 compares 2 MB/7-cycle SRAM vs. 8 MB/14-cycle STT-MRAM at -g 14 (O3CPU) and -g 10 (TimingSimpleCPU).

Kernel / Config	IPC	L2 miss rate	simSeconds	simInsts
bfs / SRAM (2 MB, 7 cyc)	0.3335	16.64%	0.43711	291 572 930
bfs / MRAM (8 MB, 14 cyc)	0.3392	7.78%	0.42985	291 572 930
sssp / SRAM (2 MB, 7 cyc)	0.3158	31.46%	0.54299	342 904 674
sssp / MRAM (8 MB, 14 cyc)	0.3313	9.50%	0.51754	342 904 674

Table 10: Part E Task 1 – gem5 O3CPU comparison at -g 14.

Task 1: Whole-Program Comparison

STT-MRAM wins on both kernels (bfs 1.017 $\times$, sssp 1.049 $\times$). Miss rates drop substantially (16.64% $\rightarrow$ 7.78% bfs, 31.46% $\rightarrow$ 9.50% sssp); each avoided miss saves $\sim$ 100+ DRAM cycles vs. only 7 extra cycles per hit.

Kernel (TimingSimple, -g 10)	SRAM simS	MRAM simS	L2 miss rate	Winner
bfs	0.020957	0.020972	27.85% $\rightarrow$ 7.78%	SRAM (+0.07%)
sssp	0.027119	0.027146	25.11% $\rightarrow$ 9.50%	SRAM (+0.10%)

Table 11: Part E – scale sensitivity at -g 10: working set fits in both caches, so capacity buys nothing and latency decides.

Task 2: Kernel Sensitivity

sssp benefits $\sim$ 3 $\times$ more (4.9% vs. 1.7%) because its 2 MB miss rate (31.46%) is much higher than bfs (16.64%), so the extra capacity captures proportionally more of its working set.

Task 3: CPU-Model Effect

The O3CPU partially hides MRAM’s 2 $\times$ hit-latency penalty by overlapping independent work during stalls. A TimingSimpleCPU exposes every stall directly, making the measured 1.7–4.9% MRAM advantage conservative for in-order cores.

Task 4: Final System Choice

Task	Result	Reasoning
2	MRAM wins at -g 14; sssp benefits more.	Poorer locality gains more from capacity.
3	O3 hides part of cache-stall cost.	In-order core exposes stalls directly.
4	MRAM wins at -g 14, SRAM at -g 10; predicted SRAM at -g 18.	Below 2 MB latency decides; at 2–8 MB capacity decides; beyond 8 MB both miss alike.

Table 12: Part E Tasks 2–4 – interpretation summary.

STT-MRAM is preferred at -g 14: the miss-rate reduction (−53% bfs, −70% sssp) outweighs the 2× hit-latency penalty. Caveats: (i) the 4× capacity ratio may be optimistic if peripherals do not scale; (ii) -g 18 was not run – overflowing both caches would equalize miss rates, letting the latency penalty dominate. The MRAM advantage is scale-dependent.

Conclusion

SRAM is faster per-access, but STT-MRAM wins at the system level when its $\sim 4\times$ density converts to large miss-rate reduction (-g 14: +1.7–4.9%). When the working set fits both caches (-g 10) or overflows both (predicted -g 18), SRAM’s 2× faster hits win. Device speed alone does not decide; workload locality, DRAM policy, and CPU model tilt the balance. Code: https://github.com/DevAbhinav-23/AMCAS_Assignments.